

Chapter 4: Fractions and Percentages

Three-week revision booklet · 6 worksheets · 180 questions

NSW syllabus outcome

MA4-FRC-C-01 — represents and operates with fractions, decimals and percentages · MAO-WM-01

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How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Fractions and Percentages topic test. Each sheet has the same shape: **30 questions**, split into ● **Easy**, ● **Medium** and ● **Challenging**. Work from the top of the page down — the questions get harder as you go. There is a ruled line under the medium and challenging questions — use it to set out your working. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Fraction basics
1	2	Adding and subtracting fractions
2	3	Multiplying and dividing fractions
2	4	Fractions, decimals and percentages
3	5	Percentages of quantities
3	6	Mixed revision — all skills

Progress tracker

Worksheet	Date completed	Score / 30	Checked by
1 Fraction basics			
2 Adding and subtracting fractions			
3 Multiplying and dividing fractions			
4 Fractions, decimals and percentages			
5 Percentages of quantities			
6 Mixed revision — all skills			

Worksheet 1: Fraction basics

Week 1 • Worksheet 1 of 6 • 30 questions

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To simplify, divide the top and the bottom by the same number. An improper fraction has a top bigger than its bottom.

EASY

15 QUESTIONS

Simplify each fraction.

1. $\frac{6}{8}$ _____

2. $\frac{4}{10}$ _____

3. $\frac{10}{15}$ _____

4. $\frac{12}{16}$ _____

5. $\frac{9}{12}$ _____

6. $\frac{8}{20}$ _____

Write each improper fraction as a mixed numeral.

7. $\frac{7}{2}$ _____

8. $\frac{9}{4}$ _____

9. $\frac{11}{3}$ _____

10. $\frac{13}{5}$ _____

11. $\frac{8}{3}$ _____

12. $\frac{15}{4}$ _____

Write each mixed numeral as an improper fraction.

13. $2\frac{1}{2}$ _____

14. $3\frac{1}{4}$ _____

15. $1\frac{2}{3}$ _____

MEDIUM

9 QUESTIONS

Find the missing number.

16. $\frac{1}{2} = \frac{?}{8}$

17. $\frac{3}{4} = \frac{?}{12}$

18. $\frac{2}{5} = \frac{?}{10}$

19. $\frac{1}{3} = \frac{?}{9}$

20. $\frac{3}{5} = \frac{9}{?}$

21. $\frac{2}{3} = \frac{8}{?}$

Simplify each fraction.

22. $\frac{24}{36}$

23. $\frac{28}{48}$

24. $\frac{35}{45}$

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

25. Order from smallest to largest: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$

26. Order from smallest to largest: $\frac{2}{3}$, $\frac{5}{6}$, $\frac{1}{2}$

27. Which is larger, $\frac{3}{5}$ or $\frac{2}{3}$? Show how you know.

28. Write $2\frac{3}{5}$ as an improper fraction.

29. Simplify $\frac{45}{60}$ fully.

30. A pizza is cut into 8 slices and 6 are eaten. Write the fraction eaten in simplest form.

Worksheet 2: Adding and subtracting fractions

Week 1 • Worksheet 2 of 6 • 30 questions

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If the denominators are the same, just add or subtract the tops. If they are different, change one fraction so they match.

EASY

15 QUESTIONS

The denominators are the same. Evaluate each expression.

1. $\frac{3}{5} + \frac{1}{5}$ _____

2. $\frac{2}{7} + \frac{3}{7}$ _____

3. $\frac{5}{8} - \frac{1}{8}$ _____

4. $\frac{4}{9} + \frac{2}{9}$ _____

5. $\frac{7}{10} - \frac{3}{10}$ _____

6. $\frac{1}{4} + \frac{1}{4}$ _____

7. $\frac{5}{6} - \frac{2}{6}$ _____

8. $\frac{3}{11} + \frac{5}{11}$ _____

9. $\frac{9}{12} - \frac{4}{12}$ _____

One denominator divides into the other. Evaluate each expression.

10. $\frac{1}{2} + \frac{1}{4}$ _____

11. $\frac{1}{3} + \frac{1}{6}$ _____

12. $\frac{3}{4} - \frac{1}{2}$ _____

13. $\frac{5}{6} - \frac{1}{3}$ _____

14. $\frac{1}{2} + \frac{3}{8}$ _____

15. $\frac{7}{10} - \frac{1}{5}$ _____

MEDIUM

9 QUESTIONS

Evaluate each expression.

16. $\frac{1}{3} + \frac{1}{4}$ _____

17. $\frac{2}{5} + \frac{1}{2}$ _____

18. $\frac{3}{4} - \frac{2}{3}$ _____

19. $\frac{5}{6} - \frac{1}{4}$ _____

20. $\frac{2}{3} - \frac{1}{5}$ _____

21. $\frac{1}{2} + \frac{2}{5}$ _____

22. $1\frac{1}{2} + 2\frac{1}{4}$ _____

23. $2\frac{1}{3} + 1\frac{1}{3}$ _____

24. $3 - \frac{1}{2}$ _____

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

25. Santi has 16 m of timber and uses $\frac{5}{8}$ of it. What fraction is left?

26. A jug holds $1\frac{1}{2}$ L. $\frac{1}{4}$ L is poured out. How much is left?

27. Add $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$.

28. A recipe needs $\frac{1}{2}$ cup of flour and $\frac{1}{4}$ cup of sugar. Find the total.

29. Find the missing fraction: $\frac{1}{4} + \underline{\hspace{1cm}} = \frac{3}{4}$

30. Which is larger, $\frac{1}{2} + \frac{1}{4}$ or $\frac{2}{3} + \frac{1}{6}$?

Worksheet 3: Multiplying and dividing fractions

Week 2 • Worksheet 3 of 6 • 30 questions

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To multiply, multiply the tops and multiply the bottoms. To divide, flip the second fraction over and multiply.

EASY

15 QUESTIONS

Evaluate each product.

1. $\frac{1}{2} \times \frac{1}{3}$ _____

2. $\frac{2}{5} \times \frac{1}{4}$ _____

3. $\frac{3}{4} \times \frac{1}{2}$ _____

4. $\frac{1}{3} \times \frac{3}{5}$ _____

5. $\frac{2}{3} \times \frac{1}{2}$ _____

6. $\frac{3}{8} \times \frac{2}{3}$ _____

7. $\frac{1}{4} \times \frac{4}{5}$ _____

8. $\frac{5}{6} \times \frac{1}{2}$ _____

9. $\frac{2}{7} \times \frac{1}{3}$ _____

Evaluate each quotient.

10. $\frac{1}{2} \div \frac{1}{4}$ _____

11. $\frac{3}{4} \div \frac{1}{2}$ _____

12. $\frac{2}{3} \div \frac{1}{3}$ _____

13. $\frac{5}{6} \div \frac{1}{2}$ _____

14. $\frac{1}{3} \div \frac{1}{6}$ _____

15. $\frac{4}{5} \div \frac{2}{5}$ _____

MEDIUM

9 QUESTIONS

Evaluate each expression.

16. $\frac{3}{8} \times \frac{4}{9}$

17. $\frac{6}{7} \times \frac{5}{8}$

18. $\frac{2}{5} \div \frac{3}{10}$

19. $\frac{5}{6} \div \frac{5}{12}$

20. $\frac{1}{2}$ of 24

21. $\frac{3}{4}$ of 20

22. $\frac{2}{3}$ of 18

23. $\frac{3}{5}$ of 40

24. $1\frac{1}{2} \times \frac{2}{3}$

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

25. Find $\frac{3}{5}$ of 40.

26. Sam ate $\frac{2}{3}$ of a 12-slice pizza. How many slices did he eat?

27. A class of 30 has $\frac{2}{5}$ girls. How many girls are there?

28. How many $\frac{1}{4}$ -cup scoops fill a 3-cup jar?

29. A tank holds 200 L when full and is $\frac{3}{4}$ full. How much is in it?

30. A ribbon 6 m long is cut into $\frac{1}{2}$ m pieces. How many pieces?

Worksheet 4: Fractions, decimals and percentages

Week 2 • Worksheet 4 of 6 • 30 questions

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Per cent means 'out of 100'. To change a fraction to a percentage, make the bottom 100 or multiply by 100.

EASY

15 QUESTIONS

Write each percentage as a fraction in simplest form.

- | | | |
|--------------|--------------|--------------|
| 1. 50% _____ | 2. 25% _____ | 3. 10% _____ |
| 4. 75% _____ | 5. 20% _____ | 6. 40% _____ |

Write each fraction as a percentage.

- | | | |
|--------------------------|-------------------------|-------------------------|
| 7. $\frac{1}{2}$ _____ | 8. $\frac{1}{4}$ _____ | 9. $\frac{3}{4}$ _____ |
| 10. $\frac{1}{10}$ _____ | 11. $\frac{1}{5}$ _____ | 12. $\frac{3}{5}$ _____ |

Write each decimal as a percentage.

- | | | |
|---------------|----------------|---------------|
| 13. 0.5 _____ | 14. 0.25 _____ | 15. 0.7 _____ |
|---------------|----------------|---------------|

MEDIUM

9 QUESTIONS

Convert as directed.

- | | | |
|--|----------------------------------|---|
| 16. 35% to a fraction
_____ | 17. 65% to a fraction
_____ | 18. $\frac{3}{20}$ to a percentage
_____ |
| 19. $\frac{7}{8}$ to a percentage
_____ | 20. 0.4 to a percentage
_____ | 21. 0.08 to a percentage
_____ |
| 22. 62% to a decimal
_____ | 23. 9% to a decimal
_____ | 24. $\frac{3}{4}$ to a decimal
_____ |

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|--|--|
| 25. Order from smallest to largest: $\frac{1}{2}$, 0.4, 45%
_____ | 26. Order from smallest to largest: $\frac{3}{4}$, 0.7, 72%
_____ |
| 27. Which is larger, $\frac{2}{5}$ or 45%? Show how you know.
_____ | 28. Write 0.375 as a fraction in simplest form.
_____ |
| 29. A test score is 18 out of 25. Write this as a percentage.
_____ | 30. Explain why $\frac{1}{3}$ does not give an exact decimal.
_____ |

Worksheet 5: Percentages of quantities

Week 3 • Worksheet 5 of 6 • 30 questions

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10% is one tenth, 50% is one half and 25% is one quarter. Build other percentages from these.

EASY

15 QUESTIONS

Find each amount.

- | | | |
|---------------------|--------------------|--------------------|
| 1. 10% of 50 _____ | 2. 50% of 80 _____ | 3. 25% of 40 _____ |
| 4. 10% of 90 _____ | 5. 50% of 24 _____ | 6. 25% of 80 _____ |
| 7. 10% of 200 _____ | 8. 50% of 18 _____ | 9. 25% of 60 _____ |

Find each amount.

- | | | |
|---------------------|---------------------|---------------------|
| 10. 20% of 60 _____ | 11. 75% of 40 _____ | 12. 30% of 50 _____ |
| 13. 5% of 200 _____ | 14. 40% of 25 _____ | 15. 60% of 30 _____ |

MEDIUM

9 QUESTIONS

Find each amount.

- | | | |
|---------------------------------|---------------------------------|---------------------------------|
| 16. 15% of 40
..... | 17. 12% of 50
..... | 18. 35% of 200
..... |
| 19. 45% of 60
..... | 20. 80% of 35
..... | 21. 90% of 70
..... |
| 22. Increase 80 by 25%
..... | 23. Decrease 80 by 25%
..... | 24. Increase 60 by 10%
..... |

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 25. A jacket costs \$80 and is discounted by 25%. Find the sale price.
.....
..... | 26. A \$60 item rises in price by 10%. Find the new price.
.....
..... |
| 27. A \$545 fridge is discounted by 40%. Find the sale price.
.....
..... | 28. A test score is 21 out of 25. Write this as a percentage.
.....
..... |
| 29. 50% of a number is 18. Find the number.
.....
..... | 30. 25% of a number is 15. Find the number.
.....
..... |

Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • 30 questions

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This sheet mixes every skill from the Fractions and Percentages chapter. Work through it as though it were the topic test.

EASY

15 QUESTIONS

Evaluate or simplify.

- | | | |
|---|--------------------------------------|---|
| 1. $\frac{6}{8}$ _____ | 2. $\frac{9}{4}$ _____ | 3. $\frac{3}{5} + \frac{1}{5}$ _____ |
| 4. $\frac{5}{8} - \frac{1}{8}$ _____ | 5. $\frac{1}{2} + \frac{1}{4}$ _____ | 6. $\frac{1}{2} \times \frac{1}{3}$ _____ |
| 7. $\frac{3}{4} \div \frac{1}{2}$ _____ | 8. $\frac{1}{2}$ of 24 _____ | 9. 50% of 80 _____ |

Convert as directed.

- | | | |
|-----------------------------|---|-------------------------------|
| 10. 25% to a fraction _____ | 11. $\frac{1}{4}$ to a percentage _____ | 12. 0.5 to a percentage _____ |
| 13. 10% of 90 _____ | 14. 20% of 60 _____ | 15. 75% of 40 _____ |

MEDIUM

9 QUESTIONS

Answer each question.

- | | | |
|---|---|--|
| 16. $\frac{1}{3} + \frac{1}{4}$ _____ | 17. $1\frac{1}{2} + 2\frac{1}{4}$ _____ | 18. $\frac{3}{8} \times \frac{4}{9}$ _____ |
| 19. $\frac{2}{5} \div \frac{3}{10}$ _____ | 20. $\frac{3}{5}$ of 40 _____ | 21. 35% to a fraction _____ |
| 22. $\frac{3}{20}$ to a percentage _____ | 23. 15% of 40 _____ | 24. Increase 80 by 25% _____ |

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 25. Order from smallest to largest: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$
_____ | 26. Sam ate $\frac{2}{3}$ of a 12-slice pizza. How many slices did he eat?
_____ |
| 27. A jacket priced at \$80 is discounted by 25%. Find the sale price.
_____ | 28. A test score is 18 out of 25. Write this as a percentage.
_____ |
| 29. Santi has 16 m of timber and uses $\frac{5}{8}$ of it. What fraction is left?
_____ | 30. 50% of a number is 18. Find the number.
_____ |